



Course Code: CS 211	Course Name: Discrete Structures
Instructors: Mr. Shoaib Raza	
Student Roll No:	Section:

Time Allowed: 50 minutes.

Maximum Points: 30 points

Attempt all questions in given sequence.

Question # 1: (04 points)

Express gcd (450, 120) as a linear combination of 120 and 450.

Question # 2: (04 points)

Consider the following relation on the set of positive integers. $R = \{(x, y) \mid x \text{ and } y \text{ have the same prime divisors}\}$
Prove or disprove that the above relation is an Equivalence relation.

Question # 3: (04 points)

Encrypt the message NEED HELP by translating the letters into numbers (A=0, B=1. . . Z=25), applying the encryption function $f(p) = (3p + 7) \bmod 26$, and then translating the numbers back into letters.

Question # 4: (04 points)

Solve the linear congruence $15x \equiv 31 \pmod{47}$.

Question # 5: (04 points)

Determine a relation on $\{1, 2, 3\}$ that is reflexive and transitive, but not symmetric.

Question # 6: (04 points)

Suppose that a computer has only the memory locations 0, 1, 2. . . 19. Use the hashing function h where $h(x) = (x + 5) \bmod 20$ to determine the memory locations in which 57, 32, and 97 are stored.

Question # 7: (06 points)

An old woman goes to market and a horse steps on her basket and crushes the eggs. The horse rider offers to pay for the damages and asks her how many eggs she had brought. She does not remember the exact number, but when she had taken them out three at a time, there were 1 egg left. When she took them four at a time, there was one egg left and when she had taken them out five at a time, then too there was one egg left. What is the smallest number of eggs she could have had?

BEST OF LUCK!